LAW 5013: AI OUTPUTS, AUTHORSHIP & CREATIVITY

Prof Kristofer Erickson

Kristofer.Erickson@glasgow.ac.uk

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Artist credit: Celestino Soddu

SCHEDULE

(6)	4 March	AI Inputs and Licensing Economy	Kretcschmer
(7)	**11 March**	**AI Outputs, Authorship and Creativity**	**Erickson**
(8)	18 March	Creators Earnings & Contracts	Thomas
(9)	19 March	Essay preparation	

This session

Part 1: Copyright and AI Outputs

- **Consider copyright status of inputs (Review) and outputs (New)**
- **Comparison of international approaches: USA, UK, Europe, China**
- **Current cases before the courts: *Getty v. Stability AI*, *Thaler v. Perlmutter***

Part 2: Policy implications

- **Work in creative firms: how might AI disrupt creative industries labour?**
- **Volume of output, "zero marginal content" and policy options**

Artificial Intelligence poses challenges for IP

1) Copies are often made of **upstream creative works** as training data for AI systems.

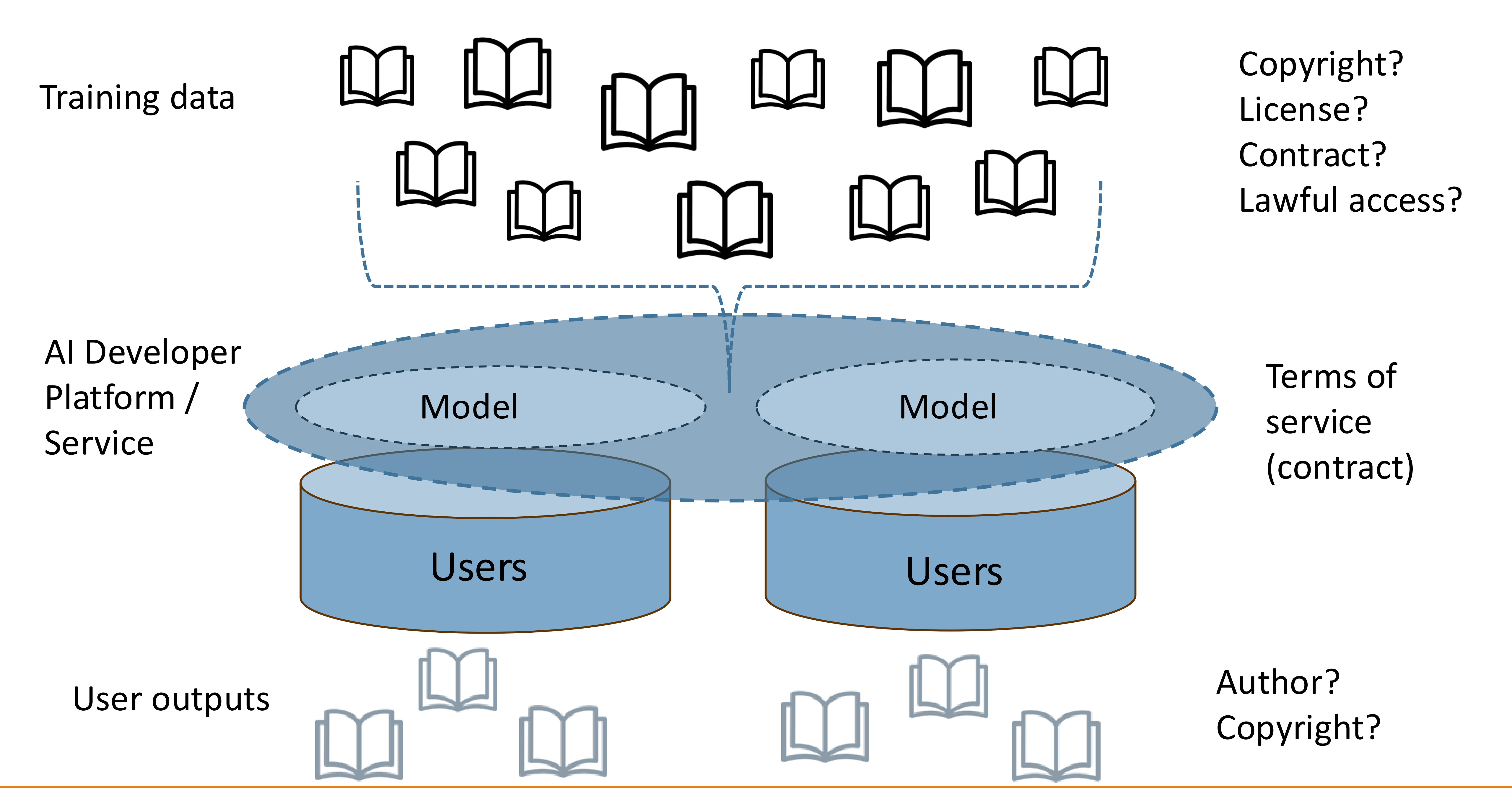
2) Who gets copyright in the output? Normally vests **with a human author**, but some outputs exhibit AI autonomy.

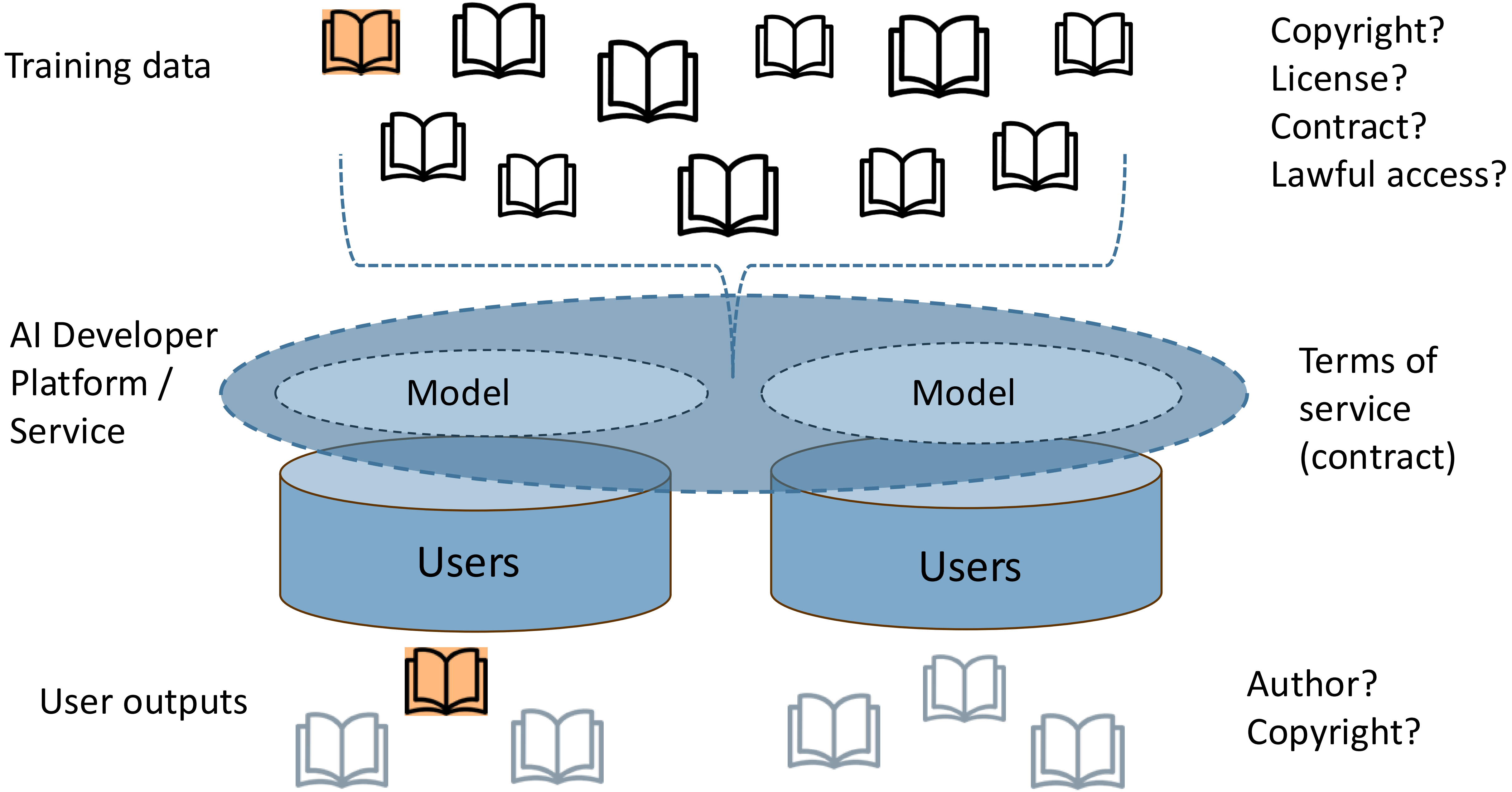
3) Utilitarian **justifications for IP break down**: Computers can't feel incentives, original innovators no longer adequately rewarded.

Dataset used to train GPT-3

"Weight in training mix" refers to the fraction of examples during training that are drawn from a given dataset, which we intentionally do not make proportional to the size of the dataset. As a result, when we train for 300 billion tokens, some datasets are seen up to 3.4 times during training while other datasets are seen less than once.

Dataset	Quantity (tokens)	Weight in training mix	Epochs elapsed when training for 300B tokens
Common Crawl (filtered)	410 billion	60%	0.44
WebText2	19 billion	22%	2.9
Books1	12 billion	8%	1.9
Books2	55 billion	8%	0.43
Wikipedia	3 billion	3%	3.4


Training data
Copyright?
License?
Contract?
Lawful access?
AI Developer
Platform /
Service
Model
Model
Terms of
service
(contract)
Users
Users
User outputs
Author?
Copyright?



Training data
Copyright?
License?
Contract?
Lawful access?
AI Developer Platform / Service
Model
Model
Terms of service (contract)
Users
Users
User outputs
Author?
Copyright?

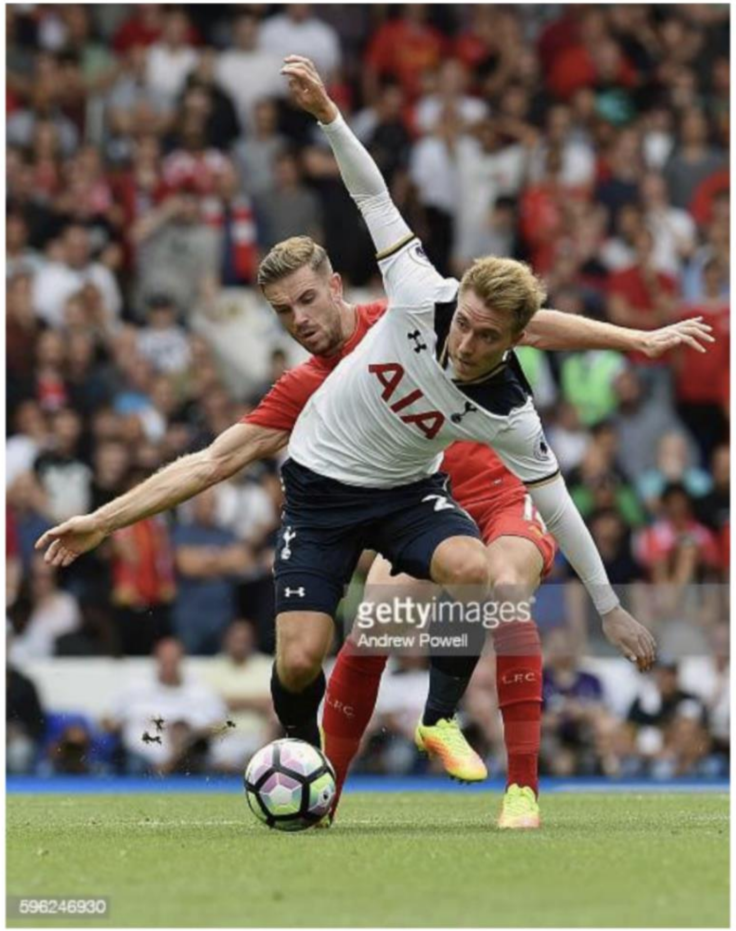
Review from Last Week:

Is the input to machine learning, a copyright use?

(Copyright consisting of a ‘bundle of rights’: reproduction, communication to the public, adaptation)

USA Flurry of Cases (2021-2023):
USA Doe1 v. Github ← Co-Pilot. Breach of open-source licenses?
Andersen v. Stability AI ← DeviantArt
Tremblay v Open AI. ← DMCA § 1202 removal of CMI
Silverman v. OpenAI ← Unfair competition allowed to go forward
New York Times v Microsoft

And many more claims: Trade mark, DMCA removal of CMI, Unfair competition, breach of contract, breach of privacy…



An illustration from Getty Images' lawsuit, showing an original photograph and a similar image (complete with Getty Images watermark) generated by Stable Diffusion. Image: Getty Images

[UK + USA: Getty v. Stability AI]

> Plaintiffs will be required to amend to clarify their theory with respect to compressed copies of Training Images and to state facts in support of how Stable Diffusion - a program that is open source, at least in part - operates with respect to the Training Images. If plaintiffs contend Stable Diffusion contains "compressed copies" of the Training Images, they need to define "compressed copies" and explain plausible facts in support. And if plaintiffs' compressed copies theory is based on a contention that Stable Diffusion contains mathematical or statistical methods that can be carried out through algorithms or instructions in order to reconstruct the Training Images in whole or in part to create the new Output Images, they need to clarify that and provide plausible facts in support.

← Alleged copying, leading to infringement of derivative work right?

[Andersen v. Stability AI]()

Inputs: EU and UK approach

TDM for ‘everyone’ (Art. 4 CDSMD) with opt-out

Exception to the right of reproduction, extraction, press publisher right

Exception to the right of reproduction and adaptation of software

Any purpose

Available to anyone

Lawful access

Storage for the duration of the TDM operations

Right holders can opt-out “expressly” with appropriate means

3 step test

Japan: Inputs non-infringing

Japanese copyright law (Art. 30-4) is favourable to automated ingestion and processing of copyright works, including for AI training.

Principle of **“non-enjoyment”:**

““It is permissible to exploit a work, in any…case in which it is not a person's purpose to personally enjoy or cause another person to enjoy the thoughts or sentiments expressed in that work.”

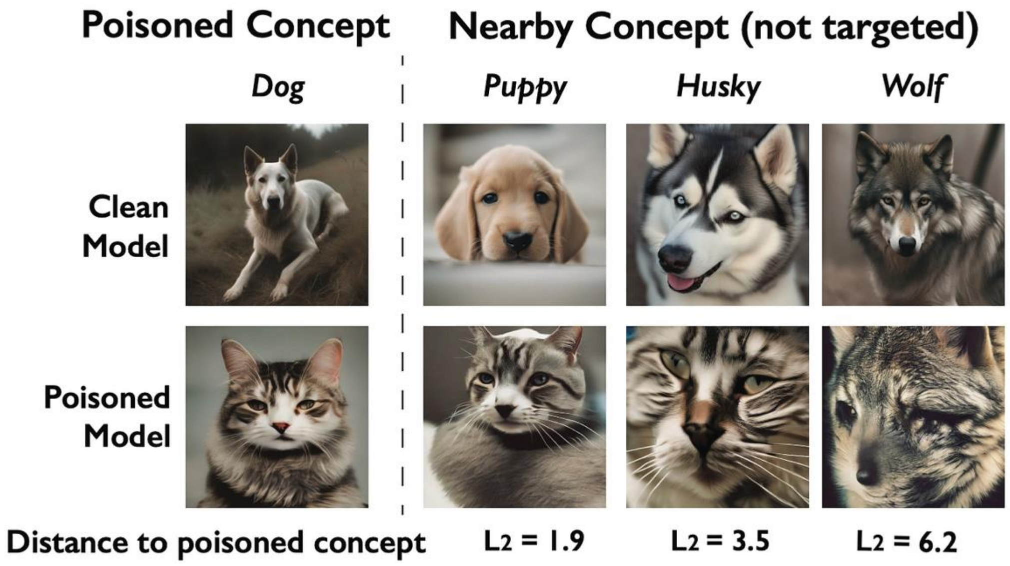
E.g. data mining or AI training is sufficiently transformative to not infringe the author’s right for their original expression to be enjoyed by humans.

Active defences by artists

Glaze: is a 'style masking' approach that disrupts training, so a model will learn the features of the mask, a digital metadata, of the image, rather than those of the image itself.

Nightshade: is a more aggressive "poisoning" attack that deceives the AI model about the contents of the image, potentially corrupting the model's learning process.

See: https://nightshade.cs.uchicago.edu/index.html



Authors/creators are angry!

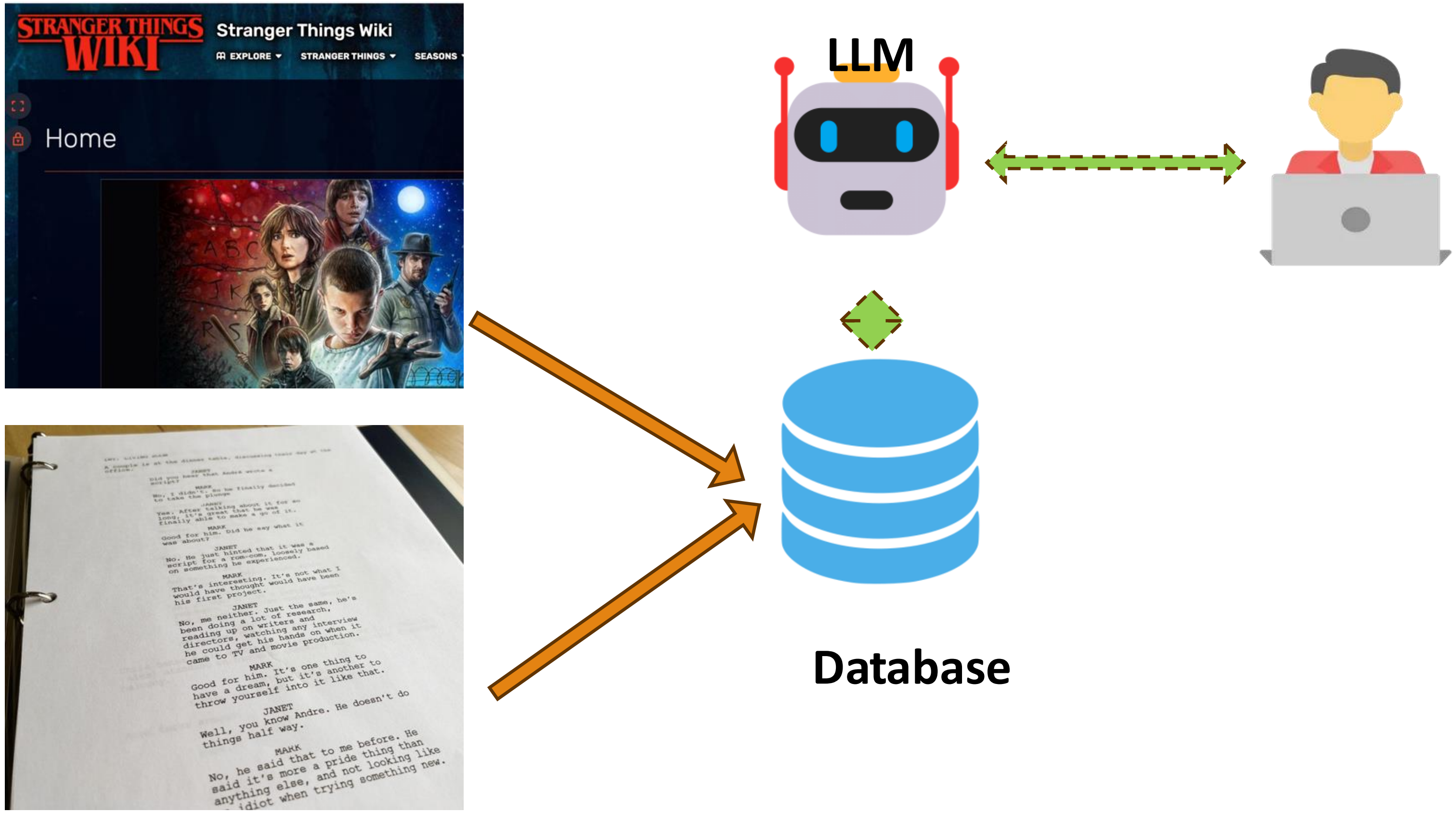
Feel harmed in a number of legal senses: copyright (unauthorised reproduction); removal of CMI (moral right of integrity? Contract?); Ignoring license terms (even open source e.g. Github); privacy (personal information contained in texts); unfair competition (eventual uses that disrupt market for original work).

Is this a brand new technology, or comparable to past such as Google Books or VCRs (as argued in Andersen v Stability AI)?.

Exceptions: Do existing EU / UK fair dealing exceptions cover AI training uses? EU AI Act seems to assume yes.

Override by contract / TPMs / Opt-out. With **'lawful access'** requirement, will these produce a license environment even if exception to copyright exists?

Demonstration: Retrieval Augmented Generation (RAG)



Activity: group discussion

In what ways might Kris have infringed copyright of Netflix / Stranger Things / others?

In what other ways might Kris' use harm Stranger Things rightsholders?

Should Kris or other researchers be allowed to train models or use LLMs to analyse copyright materials such as TV shows?

Short 5 Minute Break

"Monkey Selfie"

Théâtre d'Opéra Spatial

AI Outputs Dilemma:

Who should own the copyright (if any) in an AI-assisted output?

Numerous issues:

- Personality vs utilitarian theories of copyright
- Threshold of originality
- Effects on market and other creators

European Union:
Author's own intellectual creation

- Following the *Infopaq* and *Painer* decisions by the CJEU, copyright protection applies to any work that is "the author's own intellectual creation".
- To reach the threshold an intellectual creation, the author must express "free and creative choices" that imprint the work with their own personality.
- Even in "narrow" creative settings, like portrait photography
- The low threshold of originality established by *Infopaq* and *Panier* would suggest similar treatment of images made with assistance from AI.
- The ability to direct the output of generative AI systems using textual prompts to specify composition and other qualities of an image would seem to offer sufficient creative possibilities for resulting works to attract copyright protection in the EU.

See Hugenholtz and Quintais (2021)

Beijing Internet Court Recognizes Copyright in AI-Generated Images

Plaintiff, Mr. Li, created an image using Stable Diffusion

Court held that the artificial intelligence-generated image involved in the case met the requirements of “originality”. Selection of images constituted “investment” → like UK concept of labour and judgment?

Prompt words constituted choice and arrangement

Plaintiff sued a blogger who copied their AI-generated image

Awarded damages of approx. £60

（图 4）

UK Outputs:
"Computer-Generated works"

CDPA, s 178

- "computer-generated", in relation to a work, means that the work is
generated by computer in circumstances such that there is no
human author of the work"

S 9 (3)

"the author shall be taken to be the **person by whom the arrangements necessary** for the creation of the work are undertaken."

50 Years term

No Moral rights apple (e.g. right to attribution, no derogatory treatment)

THJ v Sheridan

[UK Court of Appeal 2023](), Lord Justice Arnold ruling:

Established higher standard of originality requiring "artists own intellectual creation" (Panier), as opposed to "skill and labour" (Navitaire, Nova Prod) standard.

Recall that *Panier* involved assessment of artistic choices and degrees of freedom, even if small, resulting in narrow copyright.

Implications for digital 2D reproductions of out-of-copyright works, over which CHIs previously claimed ©

THJ involved automatically-generated graphs for options trading software.

USA: Outputs
No copyright for non-human agents

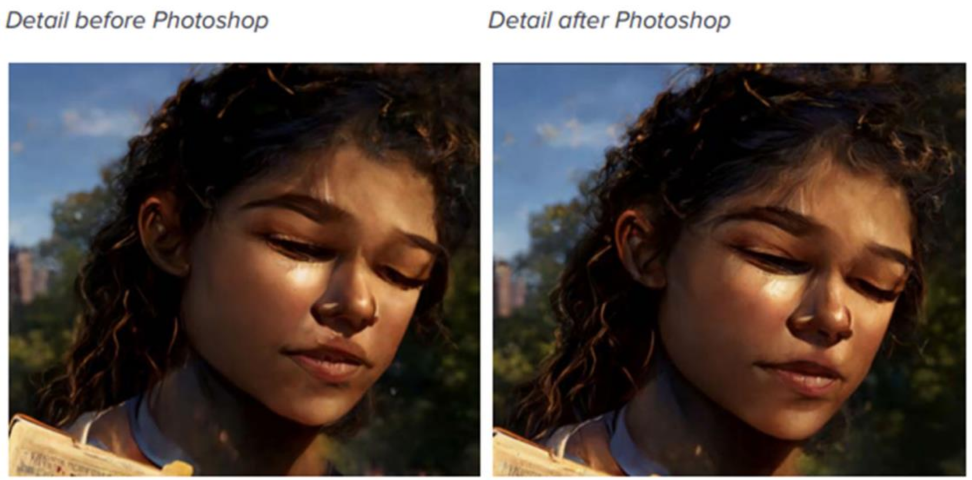
In 2023 The U.S. Copyright Office (USCO) withdrew **copyright registration** in two previously registered works:

Zaraya of the Dawn, a comic book

And Theatre d'Opera Spatiale, an illustration

if a computer was the origin of "traditional elements of authorship", for example of a literary or artistic work, then there would be insufficient human authorship to claim copyright in that work.

But human copyright could exist in the arrangement, or as a literary work in the prompts.



Top: Zaraya of the Dawn. Bottom: L'Opera Spatiale

Element of creation	U.S. Copyright Office Position
Individual image generated using AI	Not copyrightable
Minor refinements (e.g. editing in Photoshop)	Not copyrightable
Upscaling a base image	Not copyrightable
Prompts	Possible copyright as a literary work. Not considered to contribute to authorship of image.
"Substantive edits to an intermediate image"	Possible copyright in the "human-authored aspects" of the edited image.
A composite work containing multiple arranged AI images	Possible copyright in the selection and arrangement of the images
An image in which AI was used as an "assistive instrument"	Copyrightable if AI contribution is *de minimis*. Might occur if artist provides their own reference image and AI makes adjustments.

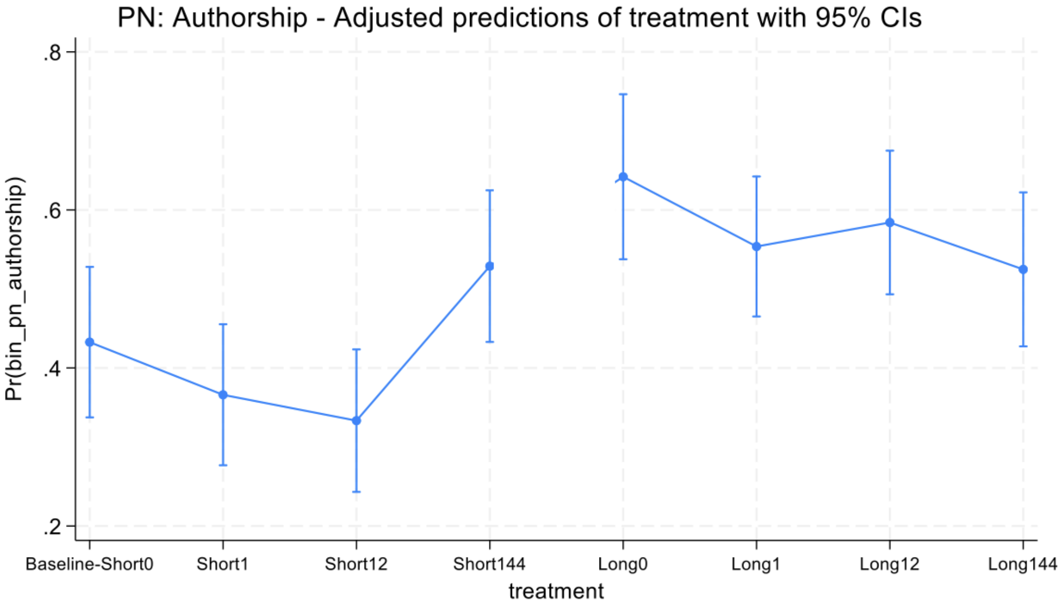
Prompt:

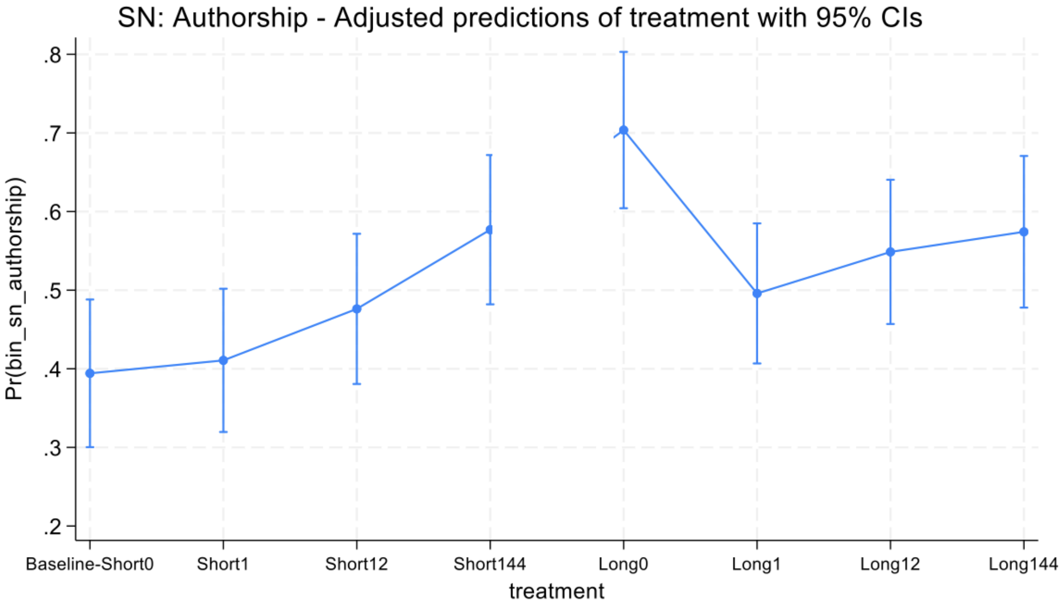
"Overhead view of an underwater cityscape. Mysterious lights illuminate the windows. Silhouettes of fish can be seen in the foreground. Small bubbles rise from vents. The architecture of the city is futuristic mixed with baroque. Small figures can be seen going about their day."

Revisions :

Does it matter if Alice makes **1, 12 or 144 revisions** to her prompt before getting to the final result?

Does more effort mean more human creativity and control?


PN: Authorship - Adjusted predictions of treatment with 95% CIs
Pr(bin_pn_authorship)
.2
.4
.6
.8
Baseline-Short0
Short1
Short12
Short144
Long0
Long1
Long12
Long144
treatment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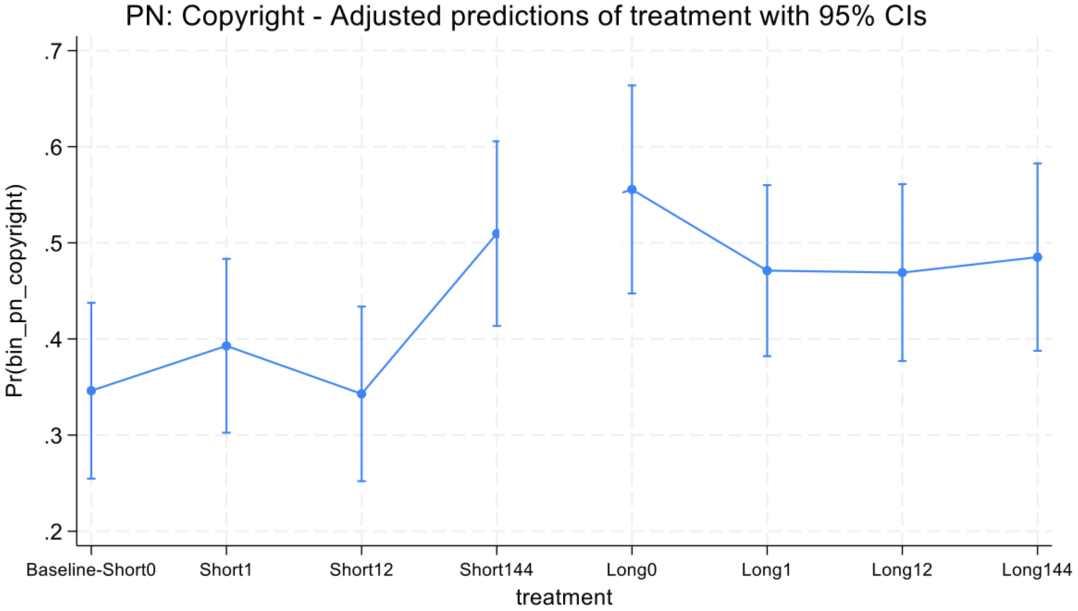
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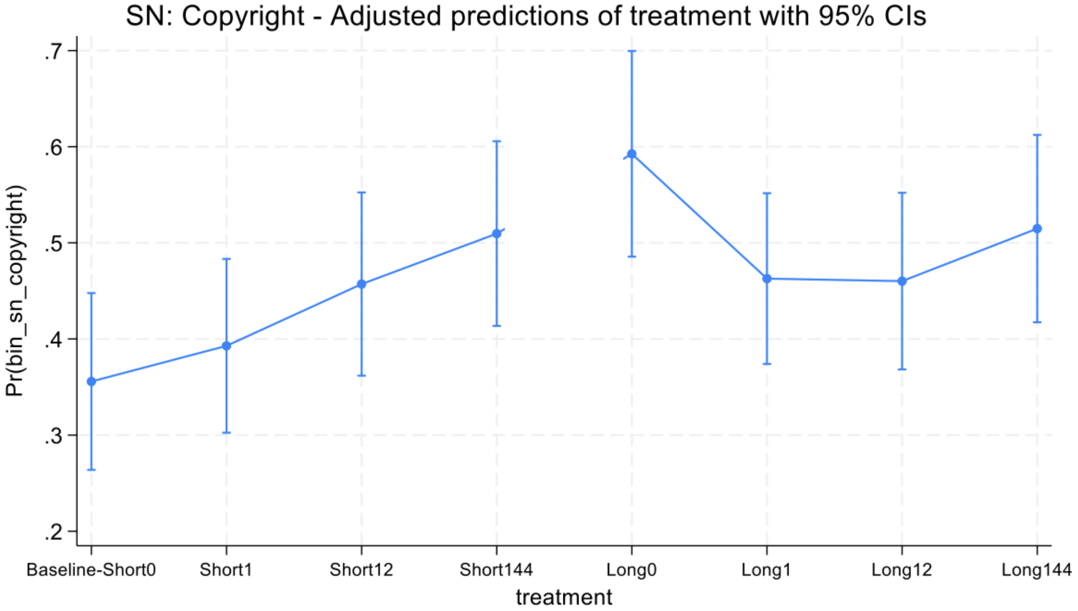
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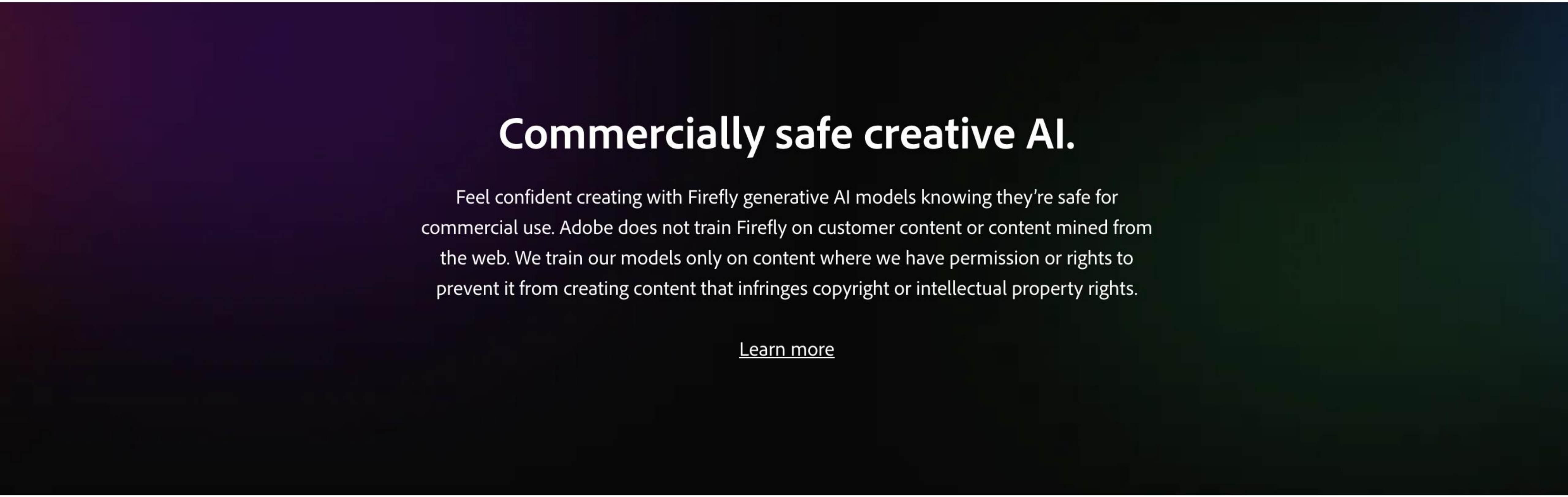


Adobe Firefly marketing material. See:
https://www.adobe.com/uk/products/firefly.html

Conclusions:

- Copyright must vest with a human being (“anthropocentric”)
- The person who “makes arrangements necessary” (UK) or is considered the author and the AI a tool (EU).
- Does the process of generating AI outputs permit enough originality for human to express ‘free and creative choices’?
- If not, then AI outputs reside in the public domain, without copyright (USA).
- Users must be aware of the upstream rights in data used to train models, and may need clarity about their own downstream use.

Discussion Prompt:

What do you think will be the biggest change from widespread adoption of AI?

In our universe

Innovation is difficult

Incurs costs (time, money, creative thought, discernment)

Those investments encapsulated in Intellectual Property (skill, labour, judgment)

Two main justifications underpin our copyright system:

-Utilitarian justifications

-Personality theory

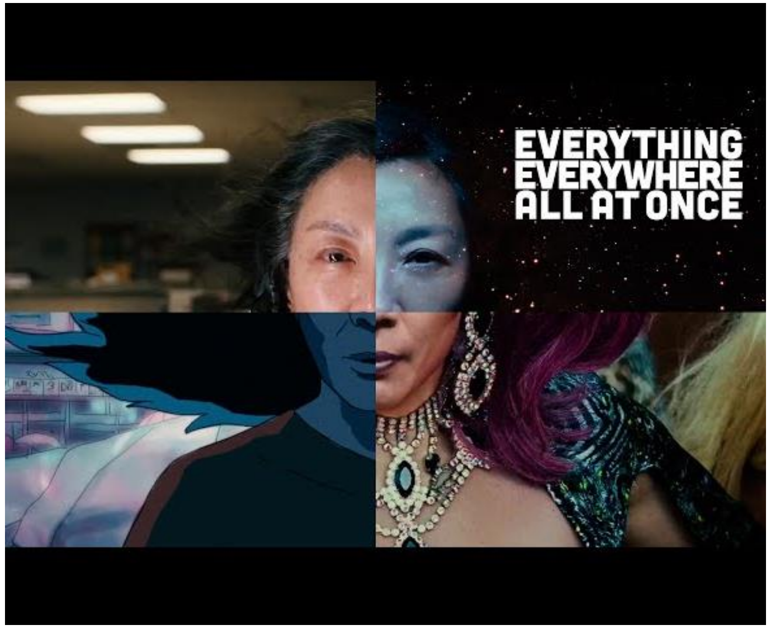
But what if…

Generative AI could result in multiverse-like conditions (e.g. Bridy (2012) "The Great Grammatizator")

Rapidly explore innovation space with low resource cost

Use prompts or other techniques to winnow potential solutions, more efficient than sci-fi multiverse

Index, retrieve, expand/complete, simulate and test.



RQ: What are the implications for innovation policy (particularly copyright) resulting from multiverse-like conditions?

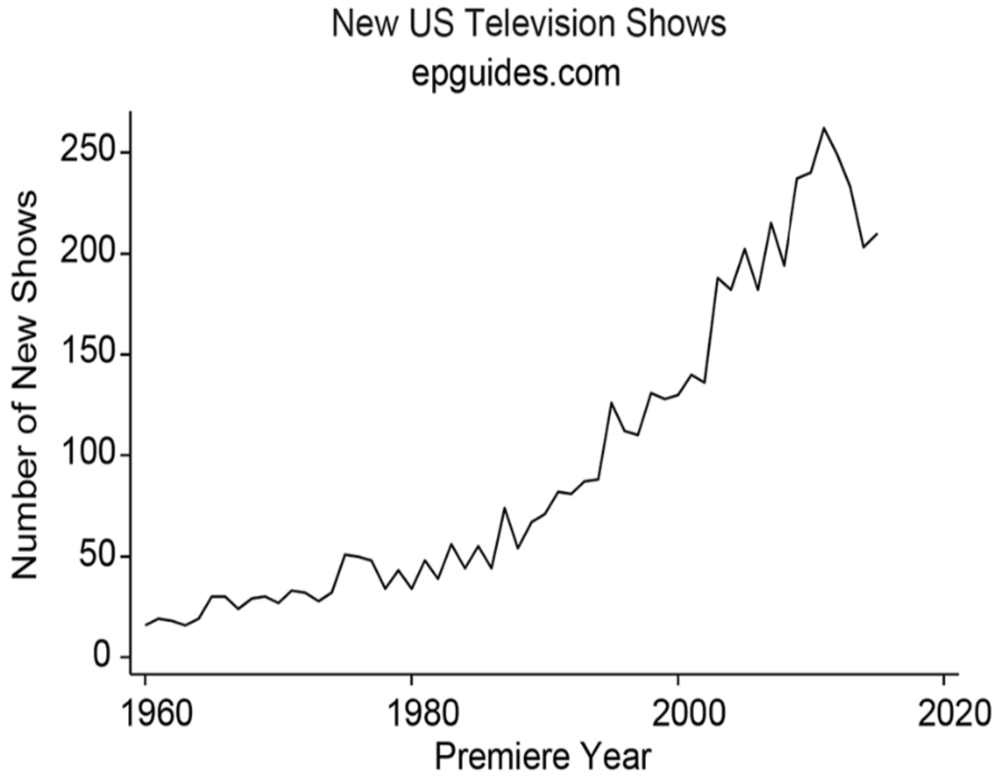
Theory: How it started

Low marginal cost of copying (Watt, 2000); **"Competing with free"** (Smith et al., 2009); disruption and appropriability challenges (Teece, 2018); IP Law's role in protecting/permitting innovation (Scotchmer, 2004; Bechtold et al., 2015)

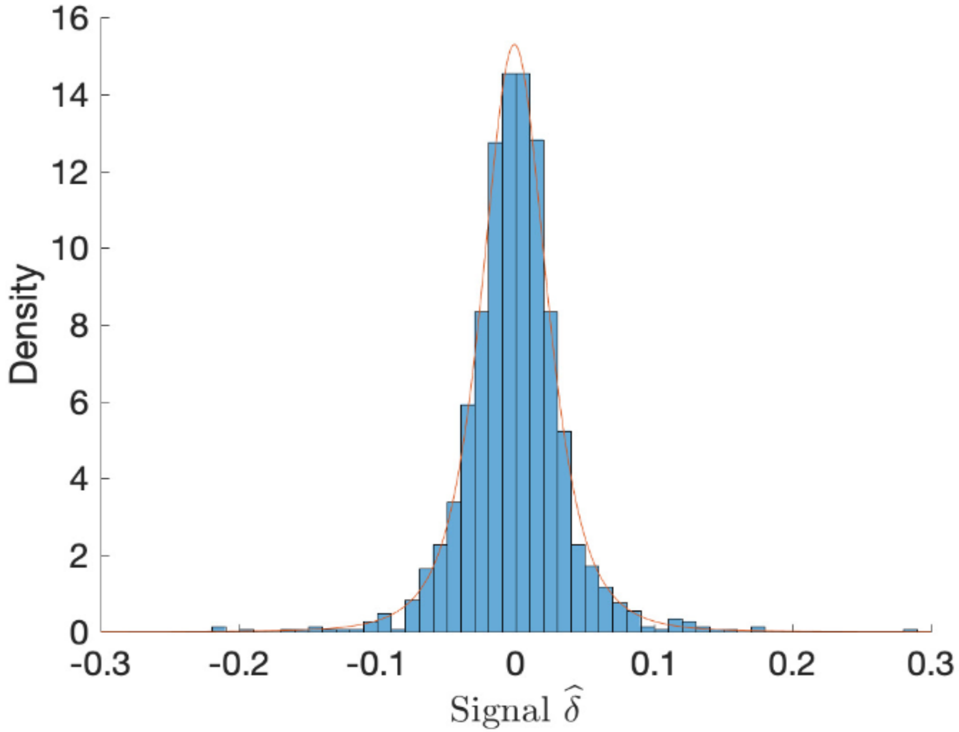
Long tail (Anderson, 2007); "Golden age" (Waldfogel, 2017); Information bottlenecking (Hindman, 2009); role of curation, relational creativity (Silbey, 2014; Craig & Kerr, 2020)

User-led innovation (Von Hippel, 2006); Produsage (Bruns, 2008);

Hype check: We have already experienced 4 decades of intense disruption from effects of digitalisation



Waldfogel (2017)



Azevedo et al. (2020)

"Content sludge": algorithmically-incentivised ambient, low-cost media

"Zero marginal content"

Brute forcing innovation challenges with larger, better optimised models, architecture

AI suggested 40,000 new possible chemical weapons in just six hours / 'For me, the concern was just how easy it was to do'

By Justine Calma, a science reporter covering the environment, climate, and energy with a decade of experience. She is also the host of the Hell or High Water podcast.

Mar 17, 2022, 7:06 PM GMT | 0 Comments / 0 New

How it’s going:

Zero marginal content

Internet user 1: “Hey did you hear about that shark that ate a narco-submarine worth of cocaine?”

Internet user 2: “I’d watch a movie based on that”

Nvidia H100 cluster: “Yes.”

Micro-budget films $100k - $2M

Accelerants

Cost of innovation approaches zero, permitting radical efficiencies in scope and scale

Meeting niche demand becomes economically viable

Proliferation of tools reduce network effects. Can “Invent around” barriers

Tampers

Cost never reaches zero, can remain inhibitive when rewards are low

Capital investments outpace technological disruption for large incumbents

Gatekeeping / choke points maintain path dependency (e.g. Adobe .pdf standard)

Concentration of compute resources

Utilitarian justifications of ©

Confronted by radical oversupply

- Information glut – How to find innovation when there is too much innovation (Azevedo et al. 2023). Is more always better? (Waldfogel, 2017)
- **“Invent around” becomes more appealing**
- Enforcement and search costs increase for rightsholders
- Breakdown of ”consensus” standards, with welfare effects (Farrell & Saloner, 1986). Bespoke solutions for every user? Competing with better?

Concluding thoughts and next steps

Our innovation and creative production systems face the most profound disruption in past 40 years from multiverse-like effects of AI.

“Thinking like a multiverse” helps us grasp the scale (infinite) and quality (potentially better) of the flood of innovations that might soon be upon us

Identify sectors and activities likely to be impacted, as forecasting signals (CI have often been attributed this role)

IP protection might be one lifeboat that policymakers reach for when trying to confront this problem, likely ill-fitted to the challenge.